
Producing Senders from Coroutine-Native Code

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Appendix A. Bridge Implementation

Abstract

An `IoAwaitable` (P4003R0^[2]) can be wrapped as a `std::execution` sender. Awaitables returning `void` or a single value map to `set_value`. Awaitables returning `error_code` map to `set_value()` on success and `set_error(ec)` on failure - no exceptions. Awaitables returning compound I/O results - any tuple-like whose first element is `error_code` with additional elements - are rejected at compile time. The coroutine body is the translation layer: it inspects the compound result, reduces it to an `error_code`, and returns that. The bridge routes the `error_code` through the three channels without exceptions.

This paper is one of a suite of six that examines the relationship between compound I/O results and the sender three-channel model. The companion papers are P4050R0^[14], “On Task Type Diversity”; P4053R0^[7], “Sender I/O: A Constructed Comparison”; P4054R0^[13], “Two Error Models”; P4055R0^[6], “Consuming Senders from Coroutine-Native Code”; and P4058R0^[15], “The Case for Coroutines.”

Revision History

R0: March 2026 (post-Croydon mailing)

- Initial version.

1. Disclosure

The authors developed and maintain Corosio^[5] and Cappy^[3] and believe coroutine-native I/O is the correct foundation for networking in C++. The authors provide information, ask nothing, and serve at the pleasure of the chair.

Cappy^[3] is a coroutine primitives library. P4055R0^[6] showed the sender-to-awaitable direction. This paper shows the reverse. The bridge depends on Cappy^[3] and beman::execution^[4], a community implementation of `std::execution` (P2300R10^[1]). The complete implementation is in Appendix A.

The authors regard `std::execution` as an important contribution to C++ and support its standardization for the domains it serves well - GPU dispatch, heterogeneous execution, and compile-time work-graph composition among them. Nothing in this paper or its companions argues for removing, delaying, or diminishing `std::execution`. The authors' position is narrower: that networking and stream I/O present a compound-result

structure that the three-channel model was not designed to carry, and that this domain is better served by a coroutine-native facility that can coexist with senders and interoperate where the domains meet. Two models, each correct for its domain, is a stronger standard than one model asked to serve both.

2. The Bridge

`as_sender` wraps any `IoAwaitable` as a `std::execution` sender. The receiver's environment carries the I/O execution context:

```
capy::thread_pool pool;

auto sndr = capy::as_sender(capy::delay(500ms));

auto op = ex::connect(
    std::move(sndr),
    demo_receiver{
        {pool.get_executor(), std::stop_token{}}},
    &done});

ex::start(op);
```

The receiver's environment answers a `get_io_executor` query with the pool's executor. The adapter extracts it, builds an `io_env`, and feeds it to `await_suspend(h, io_env const*)`. When the awaitable completes, the bridge calls `set_value` on the receiver.

```
main thread: 4256
  starting delay...
  set_value on thread 43448
  delay completed
```

The delay ran on a pool worker. Zero allocation beyond the coroutine frame.

3. The Three-Channel Problem

The bridge works for `delay`. What about `read_some`?

`read_some` returns `io_result<size_t>` - an `(error_code, size_t)` pair. The adapter must route this through three channels. The sender model provides three completion channels: `set_value` for success, `set_error` for failure, and `set_stopped` for cancellation. Algorithms like `when_all`, `upon_error`, and `retry` key on which channel fires. P4053R0^[7] documented the trade-off: route the whole pair through `set_value` and the composition algebra is bypassed; decompose it and the byte count is destroyed on error because `set_error` carries only the `error_code`. Neither option preserves both values and retains composition. The same finding now appears inside the bridge adapter itself.

There is a floor below which compound results should not cross into the sender channel model.

4. The Abstraction Floor

The solution is to not bridge compound results directly. The adapter inspects the return type structurally and rejects any tuple-like whose first element is `error_code` with additional elements:

```
template<class IoAw>
auto as_sender(IoAw&& aw)
{
    using R = decltype(
        std::declval<std::decay_t<IoAw>&>()
            .await_resume());
    static_assert(
        !detail::is_compound_ec_result_v<
            std::decay_t<R>>,
        "as_sender does not accept awaitables "
        "whose result destructures into "
        "(error_code, ...). Wrap the "
        "operation in a task<error_code> "
        "that inspects the compound result "
        "and returns the error code.");
    return awaitable_sender<std::decay_t<IoAw>>{
        std::forward<IoAw>(aw)};
}
```

The constraint is structural, not nominal. It does not name `io_result`. It asks: does the return type have a tuple protocol, is element 0 `error_code`, and are there additional elements? This catches `io_result<size_t>`, `std::tuple<error_code, size_t>`, `std::pair<error_code, size_t>`, or any user-defined type with the same shape. `std::expected<size_t, error_code>` is not caught - it lacks the tuple protocol. The constraint targets the most common I/O result shapes.

Awaitables returning a bare `error_code` - or a single-element tuple-like whose sole element is `error_code` - are binary outcomes. The bridge routes them: `set_value()` when zero, `set_error(ec)` otherwise. No exceptions.

5. Above and Below

The bridge inspects the `await_resume` return type structurally and selects the channel mapping at compile time:

<code>await_resume</code> type	Example API	<code>tuple_size</code>	Element 0	Bridge behavior
<code>void</code>	<code>delay(500ms)</code>	N/A	N/A	<code>set_value()</code>
<code>error_code</code>	<code>task<error_code></code>	N/A	N/A	<code>set_value()</code> / <code>set_error(ec)</code>
<code>io_result<></code>	<code>stream.connect(ep)</code>	1	<code>error_code</code>	<code>set_value()</code> / <code>set_error(ec)</code>
<code>int</code> , <code>string</code> , etc.	<code>task<int></code>	N/A	N/A	<code>set_value(T)</code>
<code>io_result<size_t></code>	<code>stream.read_some(buf)</code>	2	<code>error_code</code>	rejected
<code>tuple<error_code, size_t></code>	-	2	<code>error_code</code>	rejected
<code>pair<error_code, size_t></code>	-	2	<code>error_code</code>	rejected

The first four rows are above the abstraction floor. The channels work: `when_all` cancels siblings on I/O failure, `upon_error` is reachable, `retry` fires. No exceptions.

The last three rows are below the floor. Rejected at compile time.

6. The Translation Layer

To use I/O in a sender pipeline, wrap it in a `task<error_code>` that inspects the compound result and returns the error code:

```

copy::task<std::error_code>
read_all(auto& stream, auto buf)
{
    auto [ec, n] = co_await copy::read(
        stream, buf);
    if (ec)
        co_return ec;
    // use n...
    co_return {};
}

auto sndr = copy::as_sender(
    read_all(stream, buf))
| ex::upon_error(
    [](std::error_code ec) {
        std::cout << "read failed: "
                    << ec.message() << "\n";
    });

```

The `task<error_code>` lives above the floor. The `co_await copy::read(stream, buf)` lives below it. The coroutine body is the translation layer: inspect the compound result, perform application logic, return the error code. The bridge routes it through the three channels. No exceptions.

Cost: one coroutine frame per I/O operation that crosses the sender boundary.

`as_sender(stream.read_some(buf))` is a compile error, not a silent loss of error information.

7. P3552R3 Analysis

P3552R3^[12] defines `std::execution::task<T>`, a coroutine type that is also a sender. Its completion signature is `set_value_t(T)`. When `T` is `std::pair<error_code, size_t>`, the compound result lands on the value channel. This is “just use `set_value`” (P4053R0^[7] Section 5): `upon_error` is unreachable, `when_all` does not cancel siblings on I/O failure, `retry` does not fire. The programmer who writes `task<std::pair<error_code, size_t>>` has silently opted into “just use `set_value`”:

```

std::execution::task<
    std::pair<std::error_code, std::size_t>>
read_some_task(auto& stream, auto buf)
{
    auto [ec, n] = co_await stream.read_some(
        buf);
    co_return std::pair{ec, n};
}

auto sndr = read_some_task(stream, buf)
| ex::upon_error(
    [](std::error_code ec) {
        // unreachable
    });

```

`task` is general-purpose. A `static_assert` rejecting compound `error_code` results would be too broad. The constraint belongs at a bridge point with I/O intent, not on the general-purpose coroutine type. A programmer who uses `task<pair<error_code, size_t>>` directly gets the value-channel behavior documented in P4053R0^[7] Section 5 - both values preserved, composition algebra bypassed.

P3552R3^[12] converts unhandled `set_error` to an exception via `AS-EXCEPT-PTR`. The observation is architectural: `as_sender` enforces the abstraction floor at the `ioAwaitable`-to-sender boundary. `task` does not enforce it. The programmer chooses where the floor lives.

8. `split_ec`

A sender adapter can enforce the floor inside the pipeline:

```

do_read(sock, buf)           // sender completing with
                             // set_value(error_code)
| split_ec()                 // set_value() or
                             // set_error(ec)
| ex::upon_error(
    [](std::error_code ec) {
        // reachable, no exceptions
    });

```

`split_ec` advertises both `set_value_t()` and `set_error_t(std::error_code)` and selects between them at runtime. The implementation is a receiver adapter - no type erasure, no variant sender, no allocation. The complete implementation is in `Capy`^[3].

Three enforcement points, one abstraction floor. `as_sender` enforces it at the `ioAwaitable-to-sender` boundary. `split_ec` enforces it inside the pipeline. `task` does not enforce it. The programmer chooses where the floor lives.

9. Conclusion

The compound result stays below the floor. The binary outcome crosses above it. The channels work.

10. Acknowledgments

The authors thank Dietmar Kühl for `beman::execution`^[4] and for the channel-routing enumeration in `P2762R2`^[8], Michał Dominiak, Eric Niebler, and Lewis Baker for `std::execution`, Chris Kohlhoff for identifying the partial-success problem in `P2430R0`^[9], Kirk Shoop for the completion-token heuristic analysis in `P2471R1`^[10], Fabio Fracassi for `P3570R2`^[11], Peter Dimov for the refined channel mapping, and Ville Voutilainen for reflector discussion on the abstraction floor.

References

1. `P2300R10` - “`std::execution`” (Michał Dominiak et al., 2024). <https://wg21.link/p2300r10>
2. `P4003R0` - “Coroutines for I/O” (Vinnie Falco, Steve Gerbino, Mungo Gill, 2026). <https://wg21.link/p4003r0>
3. `cppalliance/capy` - Coroutine primitives library. <https://github.com/cppalliance/capy>
4. `bemanproject/execution` - Community implementation of `std::execution`. <https://github.com/bemanproject/execution>
5. `cppalliance/corosio` - Coroutine-native networking library. <https://github.com/cppalliance/corosio>
6. `P4055R0` - “Consuming Senders from Coroutine-Native Code” (Vinnie Falco, Steve Gerbino, 2026). <https://wg21.link/p4055r0>

7. **P4053R0** - "Sender I/O: A Constructed Comparison" (Vinnie Falco, Steve Gerbino, 2026).
<https://wg21.link/p4053r0>
 8. **P2762R2** - "Sender/Receiver Interface For Networking" (Dietmar Kühl, 2023). <https://wg21.link/p2762r2>
 9. **P2430R0** - "Partial success scenarios with P2300" (Chris Kohlhoff, 2021). <https://wg21.link/p2430r0>
 10. **P2471R1** - "NetTS, ASIO and Sender Library Design Comparison" (Kirk Shoop, 2021).
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 11. **P3570R2** - "Optional variants in sender/receiver" (Fabio Fracassi, 2025). <https://wg21.link/p3570r2>
 12. **P3552R3** - "Add a Coroutine Task Type" (Dietmar Kühl, Maikel Nadolski, 2025). <https://wg21.link/p3552r3>
 13. **P4054R0** - "Two Error Models" (Vinnie Falco, 2026). <https://wg21.link/p4054r0>
 14. **P4050R0** - "On Task Type Diversity" (Vinnie Falco, 2026). <https://wg21.link/p4050r0>
 15. **P4058R0** - "The Case for Coroutines" (Vinnie Falco, 2026). <https://wg21.link/p4058r0>
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Appendix A. Bridge Implementation

```
#include <boost/capy/concept/io_awaitable.hpp>
#include <boost/capy/ex/executor_ref.hpp>
#include <boost/capy/ex/io_env.hpp>
#include <boost/capy/io_result.hpp>

#include <beman/execution/execution.hpp>

#include <concepts>
#include <coroutine>
#include <exception>
#include <stop_token>
#include <tuple>
#include <type_traits>
#include <utility>

namespace boost::capy {

struct get_io_executor_t
{
    template<class Env>
    auto operator()(
        Env const& env) const noexcept
        -> decltype(env.query(
            std::declval<
                get_io_executor_t const&>()))
    {
        return env.query(*this);
    }
};

inline constexpr get_io_executor_t
    get_io_executor{};

struct io_sender_env
{
    executor_ref io_executor;
    std::stop_token stop_token;

    auto query(
        get_io_executor_t const&)
        const noexcept -> executor_ref
    {
        return io_executor;
    }
};
```

```

    }

    auto query(
        beman::execution::get_stop_token_t
            const&) const noexcept
        -> std::stop_token
    {
        return stop_token;
    }
};

namespace detail {

template<class T, class = void>
struct has_tuple_protocol
    : std::false_type {};

template<class T>
struct has_tuple_protocol<T,
    std::void_t<
        typename std::tuple_size<T>::type,
        typename std::tuple_element<
            0, T>::type>>
    : std::true_type {};

template<class T,
    bool = has_tuple_protocol<T>::value>
struct is_ec_outcome
    : std::is_same<T, std::error_code> {};

template<class T>
struct is_ec_outcome<T, true>
    : std::bool_constant<
        std::tuple_size_v<T> == 1 &&
        std::is_same_v<
            std::tuple_element_t<0, T>,
            std::error_code>>
    {};

template<class T>
constexpr bool is_ec_outcome_v =
    std::is_same_v<T, std::error_code> ||
    is_ec_outcome<T>::value;

template<class T,
    bool = has_tuple_protocol<T>::value>
struct is_compound_ec_result

```

```

: std::false_type {}];

template<class T>
struct is_compound_ec_result<T, true>
: std::bool_constant<
    std::tuple_size_v<T> >= 2 &&
    std::is_same_v<
        std::tuple_element_t<0, T>,
        std::error_code>>
{}];

template<class T>
constexpr bool is_compound_ec_result_v =
    is_compound_ec_result<T>::value;

template<class IoAw, class Receiver>
struct bridge_task
{
    struct promise_type;
    using handle_type =
        std::coroutine_handle<promise_type>;

    struct promise_type
    {
        io_env const* env_ = nullptr;

        bridge_task get_return_object() noexcept
        {
            return bridge_task{
                handle_type::from_promise(
                    *this)};
        }

        std::suspend_always
        initial_suspend() noexcept
        {
            return {};
        }

        std::suspend_always
        final_suspend() noexcept
        {
            return {};
        }

        void return_void() noexcept {}
        void unhandled_exception() noexcept {}
    };
};

```

```

template<class A>
struct transform_awaiter
{
    std::decay_t<A>& aw_;
    promise_type* p_;

    bool await_ready() noexcept
    {
        return aw_.await_ready();
    }

    decltype(auto) await_resume()
    {
        return aw_.await_resume();
    }

    auto await_suspend(
        std::coroutine_handle<> h)
        noexcept
    {
        return aw_.await_suspend(
            h, p_->env_);
    }
};

template<class A>
auto await_transform(A&& a)
{
    return transform_awaiter<A>{
        a, this};
};

handle_type h_{};

~bridge_task()
{
    if(h_)
        h_.destroy();
}

bridge_task() noexcept = default;

bridge_task(bridge_task&& o) noexcept
    : h_(std::exchange(o.h_, {}))
{

```

```

    }

    bridge_task& operator=(
        bridge_task&& o) noexcept
    {
        if(h_)
            h_.destroy();
        h_ = std::exchange(o.h_, {});
        return *this;
    }

    bridge_task(
        bridge_task const&) = delete;
    bridge_task& operator=(
        bridge_task const&) = delete;

private:
    explicit bridge_task(
        handle_type h) noexcept
        : h_(h)
    {
    }
};

} // namespace detail

template<class IoAw>
struct awaitable_sender
{
    using sender_concept =
        beman::execution::sender_t;

    using result_type = decltype(
        std::declval<std::decay_t<IoAw>&>()
            .await_resume());

    static auto make_sigs()
    {
        if constexpr (
            std::is_void_v<result_type>)
            return beman::execution::
                completion_signatures<
                    beman::execution::
                        set_value_t(),
                    beman::execution::
                        set_error_t(
                            std::exception_ptr),

```

```

        beman::execution::
            set_stopped_t()>{};
    else if constexpr (
        detail::is_ec_outcome_v<
            result_type>)
        return beman::execution::
            completion_signatures<
                beman::execution::
                    set_value_t(),
                beman::execution::
                    set_error_t(
                        std::error_code),
                beman::execution::
                    set_error_t(
                        std::exception_ptr),
                beman::execution::
                    set_stopped_t()>{};
    else
        return beman::execution::
            completion_signatures<
                beman::execution::
                    set_value_t(result_type),
                beman::execution::
                    set_error_t(
                        std::exception_ptr),
                beman::execution::
                    set_stopped_t()>{};
}

using completion_signatures =
    decltype(make_sigs());

IoAw aw_;

template<class Receiver>
struct op_state
{
    using operation_state_concept =
        beman::execution::operation_state_t;

    IoAw aw_;
    Receiver rcvr_;
    io_env env_;
    detail::bridge_task<IoAw, Receiver>
        bridge_;

    op_state(IoAw aw, Receiver rcvr)

```

```

        : aw_(std::move(aw))
        , rcvr_(std::move(rcvr))
    {
    }

    op_state(op_state const&) = delete;
    op_state(op_state&&) = delete;
    op_state& operator=(
        op_state const&) = delete;
    op_state& operator=(
        op_state&&) = delete;

    void start() noexcept
    {
        auto renv =
            beman::execution::get_env(
                rcvr_);
        auto ex = get_io_executor(renv);

        std::stop_token st;
        if constexpr (requires {
            { renv.query(
                beman::execution::
                    get_stop_token_t{}) }
            -> std::convertible_to<
                std::stop_token>; })
        {
            st = renv.query(
                beman::execution::
                    get_stop_token_t{});
        }

        env_ = io_env{ex, st, nullptr};

        bridge_ = [] (
            IoAw aw,
            Receiver rcvr,
            std::stop_token const* st)
            -> detail::bridge_task<
                IoAw, Receiver>
        {
            try
            {
                if constexpr (
                    std::is_void_v<
                        result_type>)
                {

```



```

        co_await std::move(aw);
        if (st->stop_requested())
            beman::execution::
                set_stopped(
                    std::move(
                        rcvr));
        else
            beman::execution::
                set_value(
                    std::move(
                        rcvr));
    }
    else if constexpr (
        detail::is_ec_outcome_v<
            result_type>)
    {
        auto result =
            co_await
                std::move(aw);
        if (st->stop_requested())
        {
            beman::execution::
                set_stopped(
                    std::move(
                        rcvr));
        }
        else
        {
            std::error_code ec;
            if constexpr (
                std::is_same_v<
                    result_type,
                    std::error_code
                >)
                ec = result;
            else
                ec = get<0>(
                    result);
            if (!ec)
                beman::execution
                    ::set_value(
                        std::move(
                            rcvr));
            else
                beman::execution
                    ::set_error(
                        std::move(

```

```

        rcvr),
        ec);
    }
}
else
{
    auto result =
        co_await
            std::move(aw);
    if (st->stop_requested())
        beman::execution::
            set_stopped(
                std::move(
                    rcvr));
    else
        beman::execution::
            set_value(
                std::move(
                    rcvr),
                std::move(
                    result));
}
}
catch(...)
{
    beman::execution::
        set_error(
            std::move(rcvr),
            std::current_exception
                ());
}
}(std::move(aw_),
    std::move(rcvr_),
    &env_.stop_token);

bridge_.h_.promise().env_ =
    &env_;
bridge_.h_.resume();
}
};

template<class Receiver>
auto connect(Receiver rcvr) &&
    -> op_state<Receiver>
{
    return op_state<Receiver>(
        std::move(aw_),

```

```

        std::move(rcvr));
    }

    template<class Receiver>
    auto connect(Receiver rcvr) const&
        -> op_state<Receiver>
    {
        return op_state<Receiver>(
            aw_, std::move(rcvr));
    }
};

template<class IoAw>
auto as_sender(IoAw&& aw)
{
    using R = decltype(
        std::declval<std::decay_t<IoAw>&>()
            .await_resume());
    static_assert(
        !detail::is_compound_ec_result_v<
            std::decay_t<R>>,
        "as_sender does not accept awaitables "
        "whose result destructures into "
        "(error_code, ...). Wrap the "
        "operation in a task<error_code> "
        "that inspects the compound result "
        "and returns the error code.");
    return awaitable_sender<
        std::decay_t<IoAw>>{
            std::forward<IoAw>(aw)};
}

namespace detail {

template<class Sender>
struct split_ec_sender
{
    using sender_concept =
        beman::execution::sender_t;

    using completion_signatures =
        beman::execution::completion_signatures<
            beman::execution::set_value_t(),
            beman::execution::set_error_t(
                std::error_code),
            beman::execution::set_error_t(
                std::exception_ptr),

```

```

        beman::execution::set_stopped_t()>;

Sender sndr_;

template<class Receiver>
struct ec_receiver
{
    using receiver_concept =
        beman::execution::receiver_t;

    Receiver rcvr_;

    auto get_env() const noexcept
    {
        return beman::execution::get_env(
            rcvr_);
    }

    void set_value(
        std::error_code ec) && noexcept
    {
        if (!ec)
            beman::execution::set_value(
                std::move(rcvr_));
        else
            beman::execution::set_error(
                std::move(rcvr_), ec);
    }

    void set_value() && noexcept
    {
        beman::execution::set_value(
            std::move(rcvr_));
    }

    template<class E>
    void set_error(E&& e) && noexcept
    {
        beman::execution::set_error(
            std::move(rcvr_),
            std::forward<E>(e));
    }

    void set_stopped() && noexcept
    {
        beman::execution::set_stopped(
            std::move(rcvr_));
    }

```

```

    }
};

template<class Receiver>
struct op_state
{
    using operation_state_concept =
        beman::execution::
            operation_state_t;

    using inner_op_t = decltype(
        beman::execution::connect(
            std::declval<Sender>(),
            std::declval<
                ec_receiver<Receiver>>()));

    inner_op_t op_;

    op_state(Sender sndr, Receiver rcvr)
        : op_(beman::execution::connect(
            std::move(sndr),
            ec_receiver<Receiver>{
                std::move(rcvr)}))
    {
    }

    op_state(op_state const&) = delete;
    op_state(op_state&&) = delete;
    op_state& operator=(
        op_state const&) = delete;
    op_state& operator=(
        op_state&&) = delete;

    void start() noexcept
    {
        beman::execution::start(op_);
    }
};

template<class Receiver>
auto connect(Receiver rcvr) &&
    -> op_state<Receiver>
{
    return op_state<Receiver>(
        std::move(sndr_),
        std::move(rcvr));
}

```

```

    template<class Receiver>
    auto connect(Receiver rcvr) const&
        -> op_state<Receiver>
    {
        return op_state<Receiver>(
            sndr_, std::move(rcvr));
    }
};

} // namespace detail

template<class Sender>
auto split_ec(Sender&& sndr)
{
    return detail::split_ec_sender<
        std::decay_t<Sender>>{
            std::forward<Sender>(sndr)};
}

} // namespace boost::copy

```